

Assessing invertebrate assemblages in the subsurface zone of stream sediments (0–15 cm deep) using a hyporheic sampler

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[1] Quantitative comparisons between benthic and hyporheic invertebrate communities are crucial for understanding the biological functions of the hyporheic zone, such as storage, migrations, and exchanges of invertebrates with the surface stream. Such comparisons are still hampered by the use of different techniques adapted to each habitat (benthic versus hyporheic). This work combines two different techniques for sampling the upper layers of bed sediments (0–15 cm): the semiquantitative “Bou-Rouch” pump classically used to sample the hyporheic zone (>15 cm), and the quantitative Hess sampler commonly used to sample the benthic zone (≤15 cm), in order to evaluate the quantitative efficiency of the pump in this 0–15 cm zone. First, a Bou-Rouch sample (BR) was taken within the cylinder of a Hess inserted within the streambed, then a second sample (benthic complement, BC) was collected within the Hess after removing the pump, in order to catch all invertebrates not extracted with the pump. The BR samples collected on average 14.5% of the total abundance and about 50% of the actual richness. The large range of variation indicates that the combination of the two techniques is not valid for a quantitative evaluation of benthic communities. Contrary to expectations, the pump did not collect more interstitial and groundwater invertebrates and no differences in faunal composition between upstream and downstream riffle positions were observed. Our results do not question the use of the BR technique under standard conditions i.e., when sampling the hyporheic zone, but underline how it is crucial to know its quantitative limits.

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1. Introduction

[2] The benthic and Hyporheic Zones (HZ) are closely connected habitats, the latter largely contributing to the ecosystem services provided by rivers. Hyporheic invertebrates potentially influence hydrosystem functioning through several processes [Krause *et al.*, 2011]. Their activity (e.g., moving, feeding) may have physical effects on sediment permeability (“ecosystem engineers”) and is likely to increase hydrological exchanges at the interface, and to promote microbial activity, growth, and dispersal

[Boulton, 2000, 2007]. Invertebrates transfer matter and energy from the benthic to the HZ, contribute to biogeochemical filtration and particulate organic matter decomposition (a key process of river metabolism [Navel *et al.*, 2011]), and sustain stream secondary production. The HZ exchanges invertebrates with the benthic zone, substantially contributing to surface biodiversity (as a nursery and storage zone) and resilience (as a refuge zone [Dole-Olivier, 2011]). Hyporheic invertebrates may also enhance groundwater ecosystem services such as water purification, bioremediation, and water infiltration [Boulton *et al.*, 2007]. They may be used as indicators of water exchanges, ecosystem diversity, and health [Dole-Olivier and Marmonier, 1992a], as the distribution and role of hyporheic communities are intimately linked to hyporheic hydrology. Understanding these biological functions of the HZ is crucial and requires quantitative comparisons between benthic and hyporheic assemblages to quantify benthos migrations into the HZ. However, such comparisons are still hampered by sampling problems.

[3] Although the benthic zone and HZ are intimately connected, sampling techniques have been developed independently for each compartment (benthic versus hyporheic). Several methods have been proposed to sample benthic invertebrates [Hauer and Resh, 2006], among them

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two quantitative techniques are widely used: Surber and Hess samplers [e.g., *Arscott et al.*, 2005; *Burgherr and Ward*, 2001]. Similarly, since the beginning of the biological exploration of porous aquifers and the extensive development of studies on the HZ, several techniques have been developed, improved and intensively used for sampling hyporheic invertebrates [*Williams*, 1984; *Pospisil*, 1992; *Malard et al.*, 2002]. However, because of its low accessibility, its high inherent physical complexity and temporal variability [*Hakenkamp and Palmer*, 1992; *Palmer*, 1993], sampling the HZ is more difficult than for the benthic zone. Only true quantitative techniques such as the freezing core technique and the colonization or standpipe corer [*Fraser et al.*, 1996] provide reliable density-based data collection. Unfortunately, these techniques cannot be used in all types of substrates (e.g., the standpipe corer) [*Williams and Hynes*, 1974]; they may also mix the effects of invertebrate activity and density by collecting the most active organisms (e.g., hyporheic sampling pots) [*Wagner and Feio*, 2001], or they may alter the bed sediments, preventing within-site replication (e.g., the freezing core technique) [*Stocker and Williams*, 1972]. With some precautions, some semiquantitative techniques (e.g., pumping techniques: “Bou-Rouch” or vacuum pumps) [*Bou and Rouch*, 1967; *Boulton and Stanley*, 1995] allow quantitative comparisons among closely collected samples (time series, vertical profiles, longitudinal, or lateral variations).

[4] Currently, these pumping techniques [e.g., *Bou and Rouch*, 1967; *Boulton et al.*, 2004a; *Stubbington et al.*, 2011] are the most widely used because of their practicality and compromise they represent for addressing many ecological questions such as the use of the HZ as a refuge for benthic invertebrates, the assessment of changes in the vertical distribution of benthic invertebrates during drying and flooding, and before/after disturbance [*Dole-Olivier*, 2011]. Since pumping methods can be used in a large range of freshwater habitats (glacial and mountain streams, sandy streams, gravel-bed rivers, intermittent, permanent, large rivers, head waters, backwaters, etc.) [*Dole-Olivier et al.*, 1993; *Boulton and Stanley*, 1995; *del Rosario and Resh*, 2000; *Malard et al.*, 2003; *Datry et al.*, 2007; *Wood et al.*, 2010] and both in lentic and lotic flow conditions, a number of studies refer to these techniques [e.g., *Marmonier and Creuzé des Châtelliers*, 1991; *Dole-Olivier and Marmonier*, 1992b; *Boulton and Stanley*, 1995; *Malard et al.*, 2001; *Boulton et al.*, 2004b; *Stubbington et al.*, 2009]. Nevertheless, serious methodological problems arise when quantitative data are sought. Densities between benthic and HZ cannot often be compared because different techniques are used for sampling the two adjacent habitats [e.g., *Datry*, 2012].

[5] Pumping methods do not provide the actual density of invertebrates, which would allow, for example, calculation of the proportion of benthic organisms migrating in the HZ during a disturbance, i.e., the quantitative value of the HZ as a refuge. Evaluations of the accuracy of hyporheic sampling techniques have been exclusively based on between-method comparisons [*Fraser and Williams*, 1997; *Scarsbrook and Halliday*, 2002] which do not represent a direct measurement of the precision of the pumping method. Comparison of the number of organisms collected by the pump compared to those missed by this technique is

a unique information not yet available in the literature. Knowledge of the efficiency of pumping techniques for evaluating the absolute number of individuals in a given volume, the actual number of species and the ecological characteristics of these species is valuable data for a more quantitative use of pumping methods, especially in the 0–15 cm subsurface zone.

[6] To address this issue, we propose to combine two different techniques to sample the top 15 cm of riverbed sediments. The “Bou-Rouch” semiquantitative technique, classically used to sample invertebrates living in the HZ (i.e., the hyporheos), and the quantitative Hess sampler, commonly used to sample invertebrates living in the benthic zone (i.e., the benthos), the latter supposedly able to collect more exhaustively the fauna present in a bounded volume of the habitat. This work has the following aims: (i) to search for a practical solution to avoid using two different techniques for sampling benthic and HZ, and (ii) to measure the reliability of a pumping technique for assessing invertebrate density and diversity in the top layer of river sediments. Accordingly, our main experimental hypotheses are:

[7] 1. As the pump extracts only a small proportion of the volume investigated, it underestimates the abundance of organisms actually present in the upper layer of alluvia. We test this hypothesis by collecting, within a bounded volume of habitat, all invertebrates not extracted by pumping.

[8] 2. We also expect that species diversity and richness, calculated from invertebrates collected with the pump, are similar to those calculated from invertebrates missed by the pump. However, despite a similar species richness we predict that species lists obtained in each case will be different, indicating that the pump is species selective.

[9] 3. In this case, we propose that the selectivity of the pumping technique depends on species traits such as adult body-size (meiofauna versus macrofauna) or ecology (benthic life-style versus interstitial and groundwater life-style). We also wanted to know if pump efficiency is sensitive to spatial heterogeneity linked to local geomorphology (e.g., upstream versus downstream riffles).

2. Methods and Material

2.1. Study Site

[10] Experiments took place on a braided stream, the Drôme River, a tributary of the East side of the Rhône River system. The study site is situated 10 km from the confluence with the Rhône River, in the downstream reach of a braided section with a large floodplain (~350 m wide) belonging to a Nature Reserve (Réserve Naturelle des “Ramières du Val de Drôme”). In this sector, the mean annual discharge of the stream is $17.7 \text{ m}^3 \text{ s}^{-1}$. Samples were collected during a low flow period (26–27 June 2009, discharge = $4 \text{ m}^3 \text{ s}^{-1}$) in active channels with a substrate consisting of cobbles, pebbles, and granules with very coarse sands [*Wentworth*, 1922].

2.2. Techniques

[11] To compare samples collected in the HZ with benthic samples, benthic and hyporheic samples were taken using a single technique, the Bou-Rouch pump. The pump is directly connected on a steel standpipe (20 mm internal

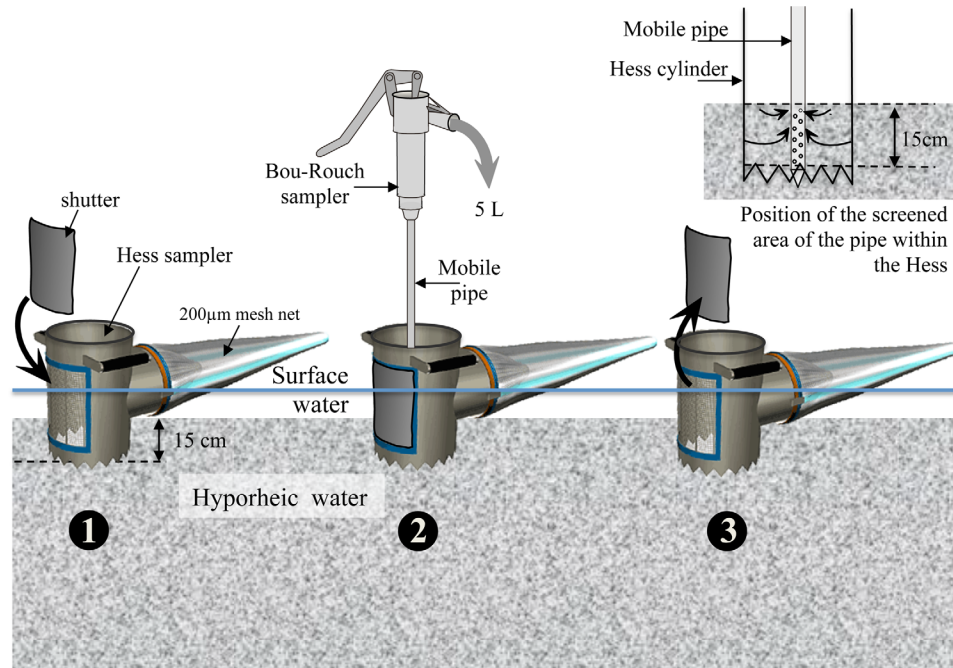


Figure 1. Sampling sequence. 1—Sealing the window of the Hess sampler with a plastic shutter, and installation of the cylinder at a 15 cm depth within the streambed. 2—Insertion of the Bou-Rouch pipe at the center of the Hess cylinder (–15 cm) and pumping of a 5 L volume of sample (BR sample). 3—Removing of the Bou-Rouch sampler, removing of the plastic shutter, and sampling of the remaining fauna with the Hess sampler (Benthic Complement, BC).

diameter, with a 13 cm long screened area consisting of 36 openings of 5 mm diameter). This pumping is designed to collect biota from a sediment layer ranging from 0 to 15 cm. As this technique is suspected to undercollect invertebrates, it was decided to collect the fauna missed by the pump using a Hess sampler. For this, we pumped the Bou-Rouch sample within a volume of bed-sediments bounded by the Hess cylinder, in order to avoid (i) collecting preferentially the fauna living in surface water and in free water (the first rows of holes being very close to the surface) and (ii) to explore the same volume of bed-sediments with the two techniques.

[12] More specifically, sampling consisted of taking two complementary subsamples in the Hess cylinder (sampling sequence in Figure 1). After pushing the cylinder of a Hess sampler to a 15 cm depth within the streambed, the first subsample was obtained by driving the standpipe at a 15 cm depth (the top of the strainer just flushing the surface of the sediment or slightly below, and the distal end of the pipe reaching a 15 cm depth) and extracting the interstitial volume isolated within the space defined by the cylinder of the Hess (i.e., 5 L) by pumping. After this, the mobile pipe and the pump were removed, but not the Hess. The second subsample, called the “benthic complement,” was taken in a standard way, in order to collect as exhaustively as possible, the fauna missed by the pump within the cylinder. A standard Hess sampling consists of removing the shutter facing the water flow (Figure 1) and washing the sediment contained in the cylinder by moving the sediment particles under the current generated (Figure 1). The finest grains and the fauna dislodged are thus collected into the downstream net, and the larger sediment particles (pebbles, cob-

bles) are removed and carefully brushed in order to collect the associated organisms, thus catching (almost) all the invertebrates contained in the cylinder. The total interstitial volume bounded by the Hess cylinder is theoretically extracted by collecting 5 L of sample (i.e., the standard Bou-Rouch sample-size). To this end, we did manufacture a cylinder, which size took into account the average porosity of the site (20–30%) and the height of the cylinder (15 cm). The resulting Hess cylinder had a diameter close to 40 cm, which isolated a volume of sediment close to 19 L. We assumed that this procedure would provide an adequate evaluation of the fauna missed by the Bou-Rouch sampler within the volume of streambed isolated by the Hess cylinder. In the sampling design, we have deliberately chosen not to maximize the differences related to the substrate (e.g., large grain sizes compared to sands), in order to have similar sediment porosities, and thus be sure to sample a same volume of habitat by pumping a sample of 5 L.

[13] Direct evaluations of sediment permeability by measuring the infiltration rate of 1 L of water in the Bou-Rouch pipe at all sampling points, did not detect any measurable differences between the points, because infiltration was instantaneous, indicating that the sediments were highly permeable at all sampling points.

2.3. Sampling Design and Samples Processing

[14] The sampling design consisted of two different braided channels and in each channel, two contrasting geomorphological positions, i.e., upstream and downstream of riffles, situated at a distance of 2–6 m on either side of the slope break. In each position, a Bou-Rouch sample (BR) and its benthic complement (BC) were taken in the

streambed at three replicate points. This sampling was repeated at four different times (26–27 June 2009 at 0, 8, 22, and 31 h). In order to compare with standard “full-Hess” samples (HESS), 12 additional Hess samples were taken at the first sampling time (three replicates, on the two channels and for two geomorphological positions) on a 0–15 cm layer. The resulting sampling design could be summarized as follows: 2 channels $\times$ 2 geomorphological positions $\times$ 3 replicates $\times$ 4 dates = 48 Bou-Rouch samples with their 48 benthic complements + 12 full-Hess samples.

[15] The 108 resulting samples were elutriated, washed, and filtered through a 200 μ m mesh. Each sample was preserved in a >70% alcohol solution mixed with eosin to stain the invertebrates. Taxonomic resolution was to species level for most of the crustaceans and to order, family or genus for the other groups (depending on the size of individuals). In order to aid in ecological interpretations, taxa were classified according to the four ecological categories (stygoxenes, occasional hyporheos, permanent hyporheos, and stygobiont) frequently used to analyze hyporheic communities [Gibert *et al.*, 1994]. Stygoxenes and occasional hyporheos are invertebrates associated with surface water (epigean) habitats, while permanent hyporheos and stygobionts are invertebrates mostly associated with groundwater (hypogean) habitats. Taxa may be also classified according to their size at the adult stage. Meiofaunal groups correspond to species that can pass through a 1 mm mesh at the adult stage (e.g., ostracods, harpacticoids, cyclopoids, cladocerans, water mites), whereas macrofauna are species retained by a 1 mm mesh (e.g., insect larvae, oligochaetes, molluscs). In this study, the distinction between macrofauna and meiofauna is used as a proxy for organism size.

2.4. Data Analysis

[16] A taxa-rank frequency ordination (%) was used for a general description and comparison of data from the BR and the BC samples. Differences in abundances, richness, and Shannon diversity between sample-types were analyzed by Student's *t* test. To determine similarities among samples in the macroinvertebrate data, nonmetric multidimensional scaling (NMDS) [Shepard, 1962; Kruskal, 1964] was performed on a Bray-Curtis distance matrix [Bray and Curtis, 1957] generating a two-dimensional ordination.

[17] Ordination is an exploratory multivariate visualization tool that allows multidimensional relationships of samples to be examined in fewer dimensions. Because ecological data sets contain samples with taxonomic objects, each with some abundance, ordination is the standard way to visualize the similarities and differences among samples or taxa. Samples that have more similar taxonomic distributions plot closer together in the ordination space.

[18] NMDS ordination iteratively searches for a best-fit solution between the rank dissimilarity indices and the distribution of samples in a low-dimension ordination space; in general, NMDS ordinations were run with only two dimensions. This nonparametric approach is appropriate for community data which are typically nonnormally and nonlinearly distributed. A “stress” level of the ordination, or measure of the goodness of fit, was calculated for each

ordination; low stress represents a better NMDS solution [Kruskal, 1964].

[19] Differences in composition of species assemblages among sample-types were tested using analysis of similarities (ANOSIM) [Warwick *et al.*, 1990]. Kruskal-Wallis one-way analysis was applied to NMDS coordinates to detect differences in the structure of macro-invertebrate assemblages among sample-types. All tests were performed with R [R Development Core Team, 2011] using the vegan [Oksanen *et al.*, 2012] and MASS [Venables and Ripley, 2002] libraries. Differences were considered significant when $P < 0.05$.

3. Results

3.1. Abundance

[20] More than 125,600 individuals were obtained from the 108 samples. The abundances obtained with the Bou-Rouch pump (BR) and with the benthic complement (BC) were compared to the full-Hess abundances (HESS) using the same number of samples taken on the same day, i.e., 12 BR samples and 12 BC samples (Figure 2A), whereas comparisons between BR and BC samples were achieved on the two sets of 48 samples (Figure 2B). The sum of BC and BR samples were compared against the HESS samples to verify sampling consistency and accuracy (Figure 2C).

[21] For the set of 12 samples, the highest abundances were observed in the HESS samples (Figure 2A), with a total of 21,194 individuals (average: 1766 ind./sample). Lower numbers of individuals were collected with the BR pump (total: 1874 ind., average: 156 ind./sample), whereas most of the fauna were found in the BC after pumping (total: 19,833 ind., average: 1653 ind./sample). The result obtained on 12 samples (Figure 2A) is also confirmed on the two sets of 48 samples (Figure 2B). No significant differences were observed between the HESS samples (total: 21,194 ind., average: 1766 ind./sample) and the sum of abundances found in the BR and BC samples (total: 21,707 ind., average: 1809 ind./sample, Figure 2C). In terms of evaluation of invertebrate abundances of the topmost layers of the sediments, the information collected by Bou-Rouch pumping was, on average, close to 14.5% to the total (i.e., BR + BC, Table 1). The 13 dominant taxa (i.e., with higher densities) observed in the BR samples were the actual dominant taxa of the community, as demonstrated by the linear relationship observed between BR and BR + BC samples ($N = 96$, $r^2 = 0.75$, $p < 0.01$, Figure 3).

3.2. Taxa Richness and Diversity

[22] A total of 109 taxa were obtained from the 108 samples. Taxa richness was similar in the HESS and the BC samples (average: 30 and 28 taxa/sample, respectively), but significantly lower in the BR samples (average: 12 taxa/sample). This trend observed on the three sets of 12 samples was corroborated by the results obtained on the two sets of 48 samples, with an average of 29 taxa in BC samples versus 12 taxa in BR samples (Figures 2A and 2B). No significant differences in richness were observed between the HESS samples (total: 61 taxa in 12 samples, average: 34 taxa/sample) and the BR + BC samples (total: 66 taxa in 12 samples, average: 30 taxa/sample, Figure 2C). The invertebrate richness of the topmost sediment

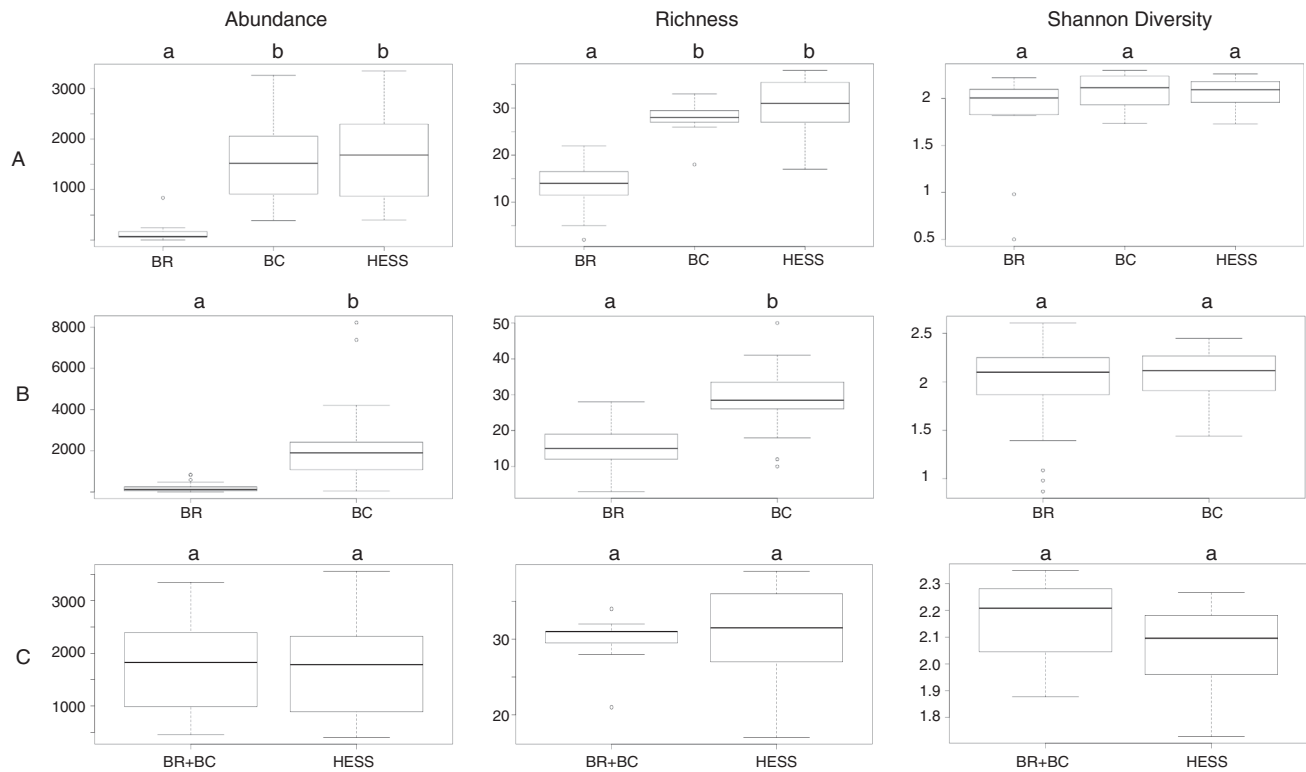


Figure 2. Total abundance, taxa richness, and Shannon diversity of invertebrates collected with the Bou-Rouch sampler (BR), the Hess sampler (HESS), and the benthic complement (BC). Comparisons based on (A) 3×12 samples, (B) 2×48 samples, and (C) 2×12 samples; extended explanations in the text. Data are represented by their median, the first and third quartiles; vertical bars are the mini/maxi values; points are outliers. For each parameter, data with the same letters (a or b) are not statistically different (t test, $p > 0.05$).

layers based on the Bou-Rouch samples was, on average, close to 50% of the total (Table 1). No significant differences in Shannon Diversity were observed between BR and BC nor between BR, BC, and HESS.

[23] Among the 109 taxa collected in the 96 BC and BR samples, 53 were common to BR and BC, whereas a total of 56 taxa were found exclusively in BR or BC samples, which represent more than half of the taxa (Table 2). A large group of taxa (43) were collected exclusively in BC, whereas a small set of taxa (13) were specific to the BR samples. Both the BC and BR collected the four ecological categories of taxa (stygoxenes, occasional hyporheos, permanent hyporheos, stygobionts; Table 2). Epigeal taxa were dominant (61 stygoxene taxa, most of them being insects larvae), stygophiles were represented by 36 taxa (23 occasional and 13 permanent hyporheos taxa), and only 12 stygobiotic taxa, mostly meiofaunal representatives, were collected. The BC samples contained more stygobionts than the BR samples (11 taxa versus 7, respectively), especially among ostracods (*Pseudocandona zschokkei*, *Cryptocandona* gr. *kieferi*, *Fabaeformiscandona breuili*, *Cavernocypris subterranea*). Only one stygobiont (the isopod *Proasellus walteri*) was exclusively found in the BR samples. An additional set of 30 epigeal taxa (stygoxenes) was collected exclusively in the BC samples (compared to nine in the BR samples). The 13 most frequently recorded taxa were collected simultaneously in BR and BC with a

quasi-identical rank frequency ordination (Figure 4). These are mostly insect larvae belonging to Ephemeroptera and Plecoptera (*Baetis* sp., *Caenis* sp., *Ecdyonurus* sp., other Heptageniidae, *Leuctra fusca*), Diptera (Orthoclaadiinae, Tanyptodinae, Chironomini, Tanytarsini, Simuliidae) and Coleoptera (*Esolus* sp.), and also the large unidentified taxa of Oligochaeta and Hydracarina. The most frequent stygobionts, *Niphargopsis casparyi*, *Niphargus* sp., *Phreatolona phreatica*, and *Dicacyclops languidoides*, found in the BR samples were generally collected with higher frequency in the BC samples.

3.3. Community Analysis

[24] A first NMDS analysis performed on the three sets of 12 samples highlights dissimilarities between the three kinds of samples (BR, BC, HESS, Figures 5a and 5b), the BR samples being isolated from BC and HESS samples (stress = 0.11). The analysis of similarities (ANOSIM) also indicates differences between the three kinds of samples (ANOSIM statistics $R = 0.41$, $p < 0.001$) with major differences between BR and BC samples, and between BR and HESS (ANOSIM statistics $R = 0.68$ and 0.72 respectively; $p < 0.001$), but no significant differences between BC and HESS samples ($R = -0.05$, $p < 0.876$). A second NMDS performed on two sets of 12 samples corresponding to (BR + BC) and HESS samples (ANOSIM statistics $R = -0.048$, $p < 0.83$) reveals no difference (data not

Table 1. Abundance (Number of Individuals) and Richness (Number of Taxa) Obtained From the 48 Bou-Rouch Samples (BR) and the 48 Benthic Complements (BC)^a

Abundance					Richness						
Sample number	BR	BC	BR + BC	BR %	Total Richness	BR	BR%	BC	Common Taxa	Taxa BR Specific	Taxa BC Specific
1	34	754	788	4.31	26	5	19.2	26	5	0	21
2	6	1151	1157	0.52	33	3	9.09	32	2	1	30
3	64	392	456	14.04	21	14	66.7	18	11	3	7
4	80	1242	1322	6.05	30	15	50	27	12	3	15
5	51	1867	1918	2.66	33	7	21.2	31	5	2	26
6	90	47	137	65.69	19	17	89.5	10	8	9	2
7	15	1061	1076	1.39	31	9	29	28	6	3	22
8	274	818	1092	25.09	32	24	75	25	17	7	8
9	240	101	341	70.38	26	24	92.3	12	10	14	2
10	76	1327	1403	5.42	32	16	50	29	13	3	16
11	223	3780	4003	5.57	42	17	40.5	41	16	1	25
12	321	182	503	68.32	26	22	84.6	19	15	7	4
13	63	3086	3149	2.00	29	12	41.4	27	10	2	17
14	1	3348	3349	0.03	31	0	0	31	0	0	31
15	166	2018	2184	7.60	32	13	40.6	30	11	2	19
16	63	1366	1429	4.41	25	12	48	24	11	1	13
17	21	3182	3203	0.66	35	6	17.1	35	6	0	29
18	62	2192	2254	2.75	35	15	42.9	33	13	2	20
19	27	2103	2130	1.27	32	11	34.4	30	9	2	21
20	31	2294	2325	1.33	37	6	16.2	37	6	0	31
21	72	2406	2478	2.91	34	14	41.2	33	13	1	20
22	134	1680	1814	7.39	25	15	60	24	14	1	10
23	244	2421	2665	9.16	39	19	48.7	36	16	3	20
24	262	2072	2334	11.23	32	15	46.9	30	13	2	17
25	251	1101	1352	18.57	31	18	58.1	28	15	3	13
26	73	2270	2343	3.12	30	15	50	28	13	2	15
27	840	1612	2452	34.26	30	22	73.3	26	18	4	8
28	238	730	968	24.59	33	23	69.7	24	14	9	10
29	590	3190	3780	15.61	42	27	64.3	38	23	4	15
30	129	572	701	18.40	25	18	72	18	11	7	7
31 ^b		7369	7369		40	0	0	40	0	0	40
32 ^b		3013	3013		26	0	0	26	0	0	26
33	311	2374	2685	11.58	37	20	54.1	34	17	3	17
34 ^b		8216	8216		49	0	0	49	0	0	49
35 ^b		2126	2126		35	0	0	35	0	0	35
36	106	610	716	14.80	30	15	50	27	12	3	15
37	178	1490	1668	10.67	31	16	51.6	28	13	3	15
38	64	1924	1988	3.22	31	11	35.5	30	10	1	20
39	134	687	821	16.32	31	17	54.8	27	13	4	14
40	346	796	1142	30.30	30	19	63.3	25	14	5	11
41	34	2238	2272	1.50	26	10	38.5	24	8	2	16
42	121	3305	3426	3.53	33	13	39.4	32	12	1	20
43 ^b		4204	4204		37	0	0	37	0	0	37
44	472	1235	1707	27.65	29	19	65.5	22	12	7	10
45 ^b		1453	1453		25	0	0	25	0	0	25
46 ^b		2438	2438		34	0	0	34	0	0	34
47	839	1967	2806	29.90	32	21	65.6	24	13	8	11
48 ^b		1276	1276		31	0	0	31	0	0	31
Total	7346	97,086	104,432			66		96	53		
Average				14.49			49.3				

^aBR % gives the percentage of specimens/taxa collected with the BR sampler (referring to the total BR + BC. "Taxa BR-specific" gives the number of species collected exclusively within the BR samples, whereas "taxa BC-specific" gives the number of species found exclusively in the BC samples. "Common taxa" are those collected simultaneously in BR and BC samples.

^bEmerged samples.

shown). A third NMDS analysis performed on the two sets of 48 samples corroborates the dissimilarities previously observed between BR and BC samples (Figure 6a, ANOSIM: $R = 0.68$, $p < 0.001$). The BR samples showed greater variability in their faunal composition than the BC samples. The taxa associated with this NMDS ordination are clearly separated along a gradient defined by the BR-specific taxa (i.e., found exclusively in BR samples) at one end, and the BC-specific taxa at the other, the taxa common

to the two kinds of samples were located between these two extremes (Figure 6b). The characterization of samples according to their geomorphological position (upstream versus downstream part of riffles) on the NMDS1 $\times$ NMDS2 ordination reveals no clear patterns linked to geomorphology (Figure 6c, ANOSIM statistics $R = -0.017$, $p < 0.125$). Species ecological classification (stygobionts, permanent hyporheos, temporary hyporheos, stygobionts, Figure 6d) or broad size classes (macrofauna or meiofauna,

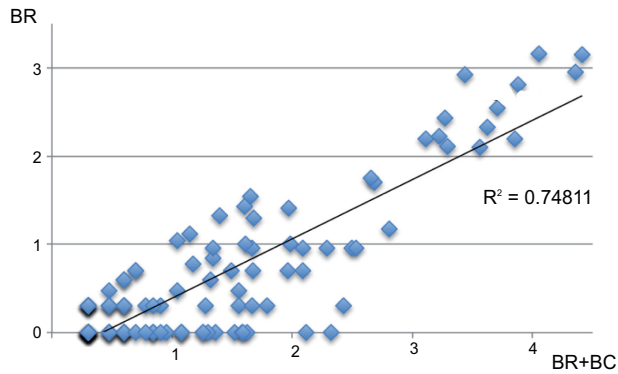


Figure 3. Abundance of the 109 taxa collected in the BR samples and in the BR + BC samples. Data expressed as $\log(x + 1)$.

Figure 6e) similarly showed no patterns linked to species ecology or size. This observation is corroborated by Kruskal-Wallis tests calculated on the coordinates of species on the first axis of NMDS (p value = 0.559 for ecological categories; p value = 0.1054 for meiofaunal versus macrofaunal species).

4. Discussion

[25] Despite the long-established research on hyporheic habitats, direct quantitative comparisons between pumping techniques and more exhaustive methods such as freeze coring [Bretschko and Klemens, 1986] or the stand pipe corer [Williams and Hynes, 1974] are still lacking. Nevertheless, some studies compared the ability of these different techniques to detect a given spatiotemporal trend [Fraser and Williams, 1997; Scarsbrook and Halliday, 2002; Descoux, 2011]. These works provided varied results depending on the observed patterns, leaving the determination of which method is most accurate and reliable unresolved, because the volumes of the alluvia employed by the techniques were not equal (e.g., 140 cm³, 1500 cm³, and 6000 cm³ for the colonization pots, the freeze core and Bou-Rouch techniques, respectively) [Scarsbrook and Halliday, 2002]. Boulton et al. [2003, 2004a] observed a non-linear relationship between sample volume and the number of taxa and individuals, indicating that hyporheic richness and abundance cannot simply be expressed per litre for comparisons among samples collected with different volumes, and *a fortiori*, for comparisons among samples collected with different techniques. In order to avoid such inconsistencies, the present study explored a priori the same volume of interstitial environment (represented by a cylinder of 40 cm in diameter and 15 cm height), measuring directly the fraction of invertebrates not collected by the pumping technique.

4.1. Assessment of Abundance in the 0–15 cm Zone

[26] Our results demonstrate that only a small fraction of the total abundance is collected by the BR pump (on average 14.5%). This result supports our first prediction that “the pumping technique underevaluates the abundance of organisms actually present in the upper layer of alluvia” and corroborates the conclusions of Fraser and Williams

[1997] and Scarsbrook and Halliday [2002] obtained from between-technique comparisons, even if in the last cases results were not directly comparable. Scarsbrook and Halliday [2002] mentioned striking differences in abundance estimates, pump samples having the fewest animals, whereas numbers in colonization pots and freeze-corings were higher and almost identical to each other. Nevertheless, the percentage of organisms collected in our BR samples was very low compared to BC samples. A bias of BR pump toward small and less tenacious animals has been pointed out by Fraser and Williams [1997] on chironomid larvae of different size classes. They related this result to hydraulic conductivity and to the size of interstitial spaces. This filtering effect was also discussed by Boulton et al. [2004a]. In our study, this bias is certainly enhanced by the position of the sampler in the subsurface area (in other studies, samples were taken in deeper zones) and the highest abundance of individuals at the sediment surface. This very low proportion of individual numbers in the BR samples may indicate that the pump has a very low efficiency for catching invertebrates living in the most superficial zone of sediments (0–15 cm). Such organisms are typically larger-bodied than those living in the interstices (e.g., late stages of insect larvae) and are often adapted to resist strong currents (authors’ personal observation). Nonsignificant differences between the BR + BC and full-Hess sets of samples confirm the reliability of the sampling protocol and demonstrate that possible external contaminations, i.e., interstitial organisms passively attracted through the bottom of the cylinder and collected later in the benthic complement, are negligible (Figure 2).

4.2. Assessment of Richness in the 0–15 cm Zone

[27] Our data indicate that the BR pump collects significantly fewer taxa than actually present in the explored sediments (about 50%), allowing us to reject our second prediction that “species diversity and richness, calculated from invertebrates collected with the pump, are similar to those calculated from invertebrates missed by the pump. However, despite a similar species richness we predict that species lists obtained in each case will be different, indicating that the pump is species-selective,” while Fraser and Williams [1997] mentioned similar qualitative performances between the BR sampler and a more exhaustive method such as freeze-coring. Nevertheless, Scarsbrook and Halliday [2002] observed that pump sampling collects only a small fraction of the animals actually present, representing 54% of the total taxa list (collected with colonization pots, freeze-coring and pumping). As taxonomic richness is underestimated by the BR pump, species lists obtained by BR and BC clearly have a different faunal composition, more than half of the taxa being collected exclusively in one of the two types of samples (BR or BC). The BR sampler has been criticized for having a strong bias toward small, less tenacious invertebrates [Scarsbrook and Halliday, 2002; Boulton et al., 2004a], indicating greater efficiency at least for sampling the typical interstitial fauna (stygo-bionts and permanent hyporheos) or the meiofauna. Contrary to expectations, the BC samples contained more stygobionts and taxa from the permanent hyporheos than the BR samples, which are dominated by epigeic taxa

Table 2. List of the 109 Taxa Identified in the 48 Bou-Rouch Samples (BR) and the 48 Benthic Complements (BC)^a

13	43	53
Taxa Specific to Bou-Rouch Sampling	Taxa Specific to Benthic Complement	Common Taxa
<i>Dugesia tigrina</i>	<i>Hydra</i>	<i>Hydracarina</i>
<i>Eulamellibranchia</i>	<i>Gyraulus</i> sp.	<i>Oligochaeta</i>
<i>Eucyclops macruroides</i>	<i>Bythiospeum</i> sp.	<i>Nematoda</i>
<i>Ectocyclops phaleratus</i>	<i>Bithynia</i> sp.	<i>Dendrocoelum</i>
<i>Asellus</i> sp.	<i>Attheyella crassa</i>	<i>Alona</i> cf. <i>phreatica</i>
<i>Proasellus walteri</i>	<i>Bryocamptus minutus</i>	<i>Diacyclops languidoides</i> s. l.
<i>Heptagenia</i> sp.	<i>Pseudocandona albicans</i>	<i>Acanthocyclops sensitivus</i>
<i>Hexatomi</i>	<i>Pseudocandona zschokkei</i>	<i>Speocyclops kieferi</i>
<i>Oulimnius</i> sp.	<i>Cryptocandona</i> gr. <i>kieferi</i>	<i>Paracyclops imminutus</i>
<i>Hydroporinae</i>	<i>Fabaeformiscandona breuili</i>	<i>Acanthocyclops</i> cf. <i>robustus</i>
<i>Odontocerum albicorne</i>	<i>Ilyocypris bradyi</i>	<i>Eucyclops serrulatus</i>
<i>Cymus</i> sp.	<i>Prionocypris zenkeri</i>	<i>Macrocyclops albidus</i>
<i>Cordulegaster</i> sp.	<i>Heterocypris reptans</i>	<i>Bryocamptus zschokkei</i>
	<i>Psychrodromus olivaceus</i>	<i>Candona candida</i>
	<i>Cavernocypris subterranea</i>	<i>Pseudocandona</i> sp. (epigean)
	<i>Gammarus roeseli</i>	<i>Cypridopsis</i> gr. <i>vidua</i>
	Perlodidae	<i>Potamocypris villosus</i>
	<i>Chloroperlidae</i>	<i>Gammarus pulex</i>
	<i>Nemoura</i> sp.	<i>Gammarus fossarum</i>
	<i>Protonemura</i> sp.	<i>Niphargus</i> sp.
	<i>Rhithrogena</i> sp.	<i>Niphargopsis casparyi</i>
	<i>Leptophlebia</i> sp.	<i>Leuctra geniculata</i>
	<i>Paraleptophlebia</i> / <i>Habroleptoides</i>	<i>Leuctra fusca</i>
	<i>Antocha</i> sp.	<i>Leuctra major</i>
	<i>Eriopterini</i>	<i>Perla</i> sp.
	<i>Hemerodromiinae</i>	<i>Marthamea</i> sp.
	<i>Rhagionidae</i>	<i>Oligoneuriella rhenana</i>
	<i>Atrichops crassipes</i>	<i>Caenis</i> sp.
	<i>Atherix</i> sp.	<i>Ecdyonurus</i> sp.
	<i>Anthomyidae</i>	Other <i>Heptageniidae</i>
	<i>Psychodidae</i>	<i>Baetis</i> spp
	<i>Stratiomyidae</i>	<i>Acentrella sinaica</i>
	<i>Blepharicera</i> sp.	<i>Serratella ignita</i>
	<i>Stenelmis</i> sp.	<i>Choroterpes picteti</i>
	<i>Laccobius</i> sp.	<i>Simuliidae</i>
	<i>Hydrocyphon</i> sp.	<i>Hexatoma</i> sp.
	<i>Yola bicarinata</i>	<i>Tabanidae</i>
	<i>Dryops</i> sp.	<i>Ceratopogonidae</i>
	<i>Psychomya pusilla</i>	<i>Orthoclaadiinae</i>
	<i>Rhyacophila</i> sp.	<i>Tanypodinae</i>
	<i>Leptoceridae</i>	<i>Tanytarsini</i>
	other <i>Polycentropodidae</i>	<i>Chironomini</i>
	<i>Gomphidae</i>	<i>Clinocerinae</i>
		<i>Tipulidae</i>
		<i>Elmis</i> sp.
		<i>Esolus</i> sp.
		<i>Limnius</i> sp.
		<i>Hydropsyche incognita</i>
		<i>Hydropsyche exocellata</i>
		<i>Hydropsyche contubernalis</i>
		<i>Cheumatopsyche lepida</i>
		<i>Hydroptila</i> sp.
		<i>Micronecta</i> sp.

^aThe taxa are classified into three groups: taxa specific to BR and BC samples respectively, and taxa common to BR and BC samples.

Stygoxene

Stygophile (occasional hyporheos)

Stygophile (permanent hyporheos)

Stygobiont

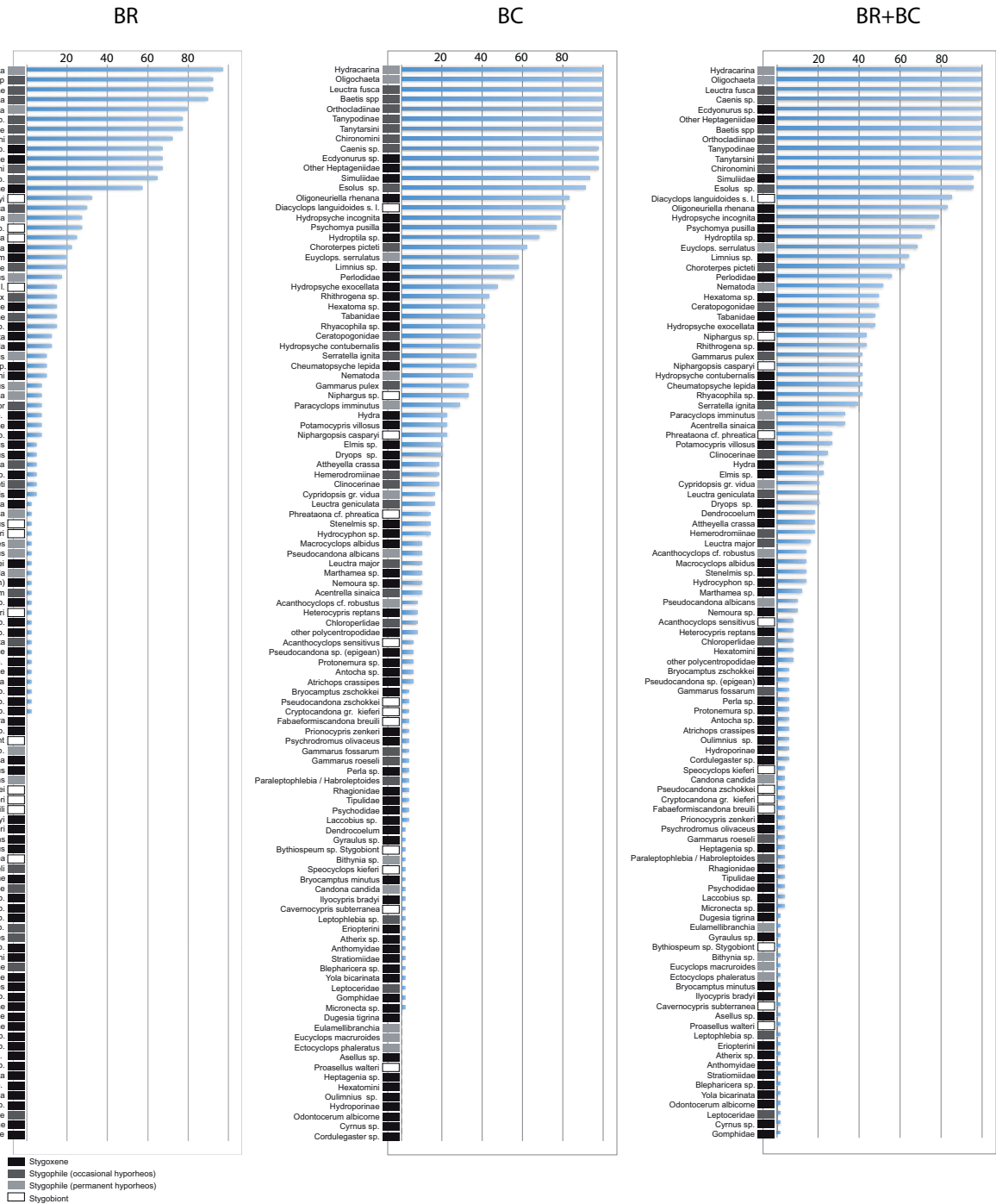


Figure 4. Taxa-rank frequency ordination in the Bou-Rouch (BR), Benthic Complement (BC), and exhaustive (i.e., BR + BC) samples.

(stygoxens and occasional hyporheos). With regard to typical meiofaunal taxa (e.g., cyclopoid and harpacticoid copepods), the BR pump estimates copepods fairly well (only two taxa absent in BR samples) but significantly underestimates ostracods (about 2/3 of taxa absent in BR samples), while neither their size, shape, nor resistance mechanisms against the current explains such differences. Indeed, compared to the macrofauna, the copepods and ostracods collected in this study have a similar size (<2 mm) and in rough approximation both are ovoid in body

shape. Moreover, they do not display behavioral adaptations or morphological structures for resistance to the pumping current (structures of attachment to the substrate, such as suckers, hooks, and special spines).

4.3. Community Analysis

[28] Our last prediction supposed that “the selectivity of pumping technique depends on species traits such as adult body-size (meiofauna versus macrofauna) or ecology (benthic life-style versus interstitial and groundwater life-

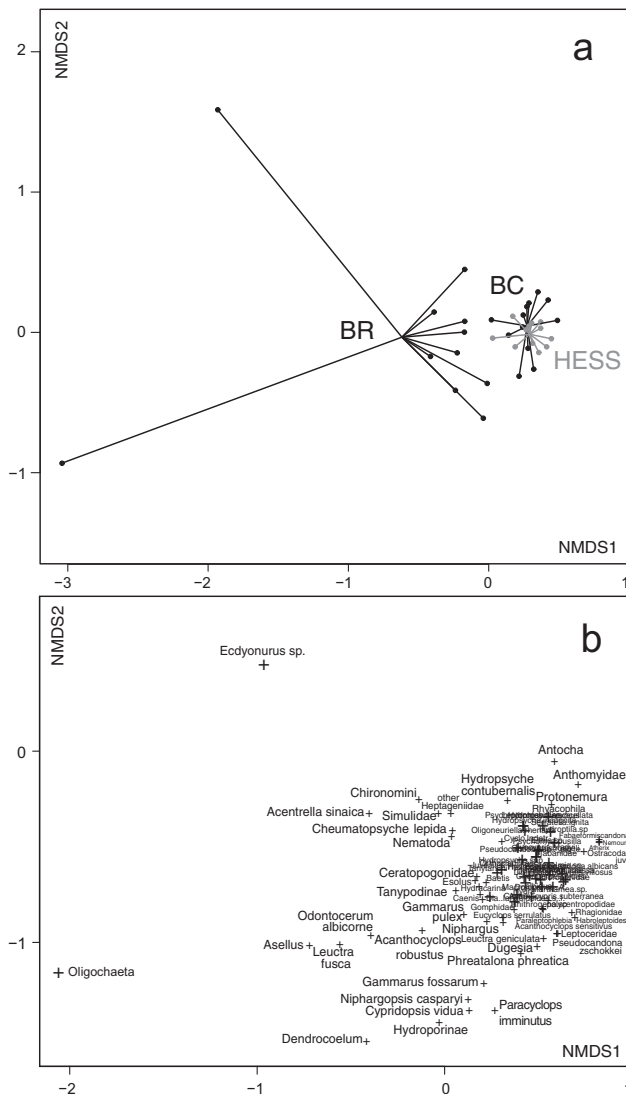


Figure 5. Results of the NMDS processed on the three sets of 12 samples: a—representation of the 36 samples, b—representation of the taxa. BR = Bou-Rouch samples, BC = benthic complement samples, and Hess = Hess samples (Stress = 0.11).

style).” The NMDS analysis demonstrates that the BC and the Hess samples are similar and provide the same description of the benthic-hypobenthic community, whereas the BR samples give another and distinct representation of faunal assemblages (Figure 5a). The contrasting assemblage compositions observed on the three sets of 12 samples (BR, BC, Hess) is verified on the two sets of 48 samples (BR, BC, Figure 6a). Although the BR and BC samples share a large number of species (Table 2, Figure 5), the number of species specific to each sample type is also very high (56 species), representing more than half of the total number of species (109). The life-style of the species (Table 2, Figure 4) does not appear to explain this strong distinction between sample-types. Contrary to the prediction, the pump does not collect more hypogean and hyporheic taxa (permanent hyporheos) than typically benthic organisms (stygoxenes, temporary hyporheos). The NMDS analysis

corroborated this result indicating no bias of the BR toward interstitial groups or small-sized meiofaunal taxa (Figures 6d and 6e). Selectivity of the pumping technique is also likely related to the resistance of each species to the current flow. Unfortunately, it was not possible to test this effect because while such information is available in the literature for the majority of the macrofauna species [Tachet et al., 2000], it is not available for the meiofauna, which represent a large number of species in our data set. Finally, we wanted to know if “pump efficiency is sensitive to spatial heterogeneity linked to local geomorphology (e.g., upstream versus downstream riffles),” because it is well recognized that the composition and distribution of hyporheic assemblages are strongly influenced by stream geomorphology at several scales [Dole-Olivier, 2011; Capderrey et al., 2013], and especially at the riffle scale [Creuzé des Châtelliers and Reygrobellet, 1990], as is also the case for benthic fauna [Armitage et al., 1974; Scullion et al., 1982; Brown and Brussock, 1991; Boulton and Lake, 1992; Carter and Fend, 2001; Davy-Bowker et al., 2006]. Different assemblages normally correspond to different species traits (e.g., regarding the resistance to water current), which may induce a bias when sampling by pumping methods. Our results did not stress such bias. First, comparison between Figures 6a and 6c indicates that the partition of samples observed between BR and BC is very different to the partition observed for upstream and downstream zones of riffles. Moreover, the distinction between upstream and downstream samples is statistically not significant (Figure 6c).

5. Conclusions

[29] This work emphasizes three major points that have not previously been reported in the literature. First, this paper reflects a search for a single sampling technique usable in the benthic and HZ since there is a crucial lack of sampling methodologies, which allow quantitative comparisons of the two habitats. Standard methods used for sampling the benthos are not usable for comparisons with the hyporheos as they are absolutely not suitable in deep areas. Thus, it was necessary to test a method derived from hyporheic techniques. Due to their ease of use in a wide range of field situations, the Bou-Rouch equipment and other derived pumping devices have been extensively and usefully used around the world and have greatly contributed to increasing our knowledge of hyporheic assemblages. Our method of pumping in the 0–15 cm inside a Hess cylinder zone, should allow quantitative comparisons with deep hyporheic samples (≥ 50 cm) taken with the BR pump alone. Unfortunately, this methodology seems unsuccessful because of several biases.

[30] The second relevant contribution was to provide direct measurements of these biases by collecting all organisms not extracted by suction. Studies questioning the bias of the BR method demonstrated the effect of sample volume [Danielopol, 1976; Hunt and Stanley, 2000; Boulton et al., 2003, 2004a; Kibichii et al., 2009] and of pumping rate [Hunt and Stanley, 2000] on the assessment of abundance and richness, or indicated a probable filtering effect of the sediments [Fraser and Williams, 1997; Boulton

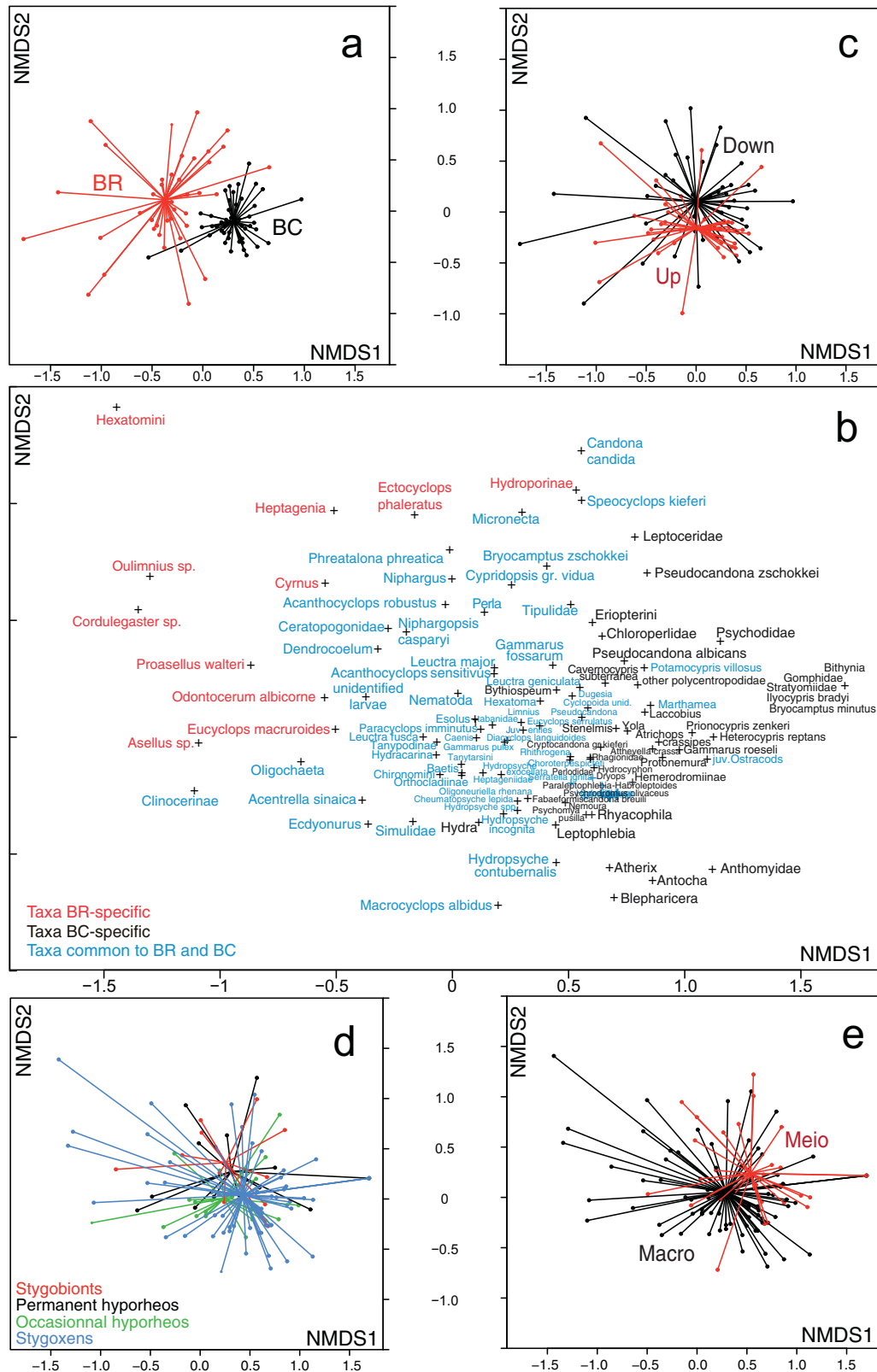


Figure 6. Results of the NMDS processed on the two sets of 48 samples: a—representation of the 96 samples according to sample-type (BR = Bou-Rouch samples, BC = benthic complement samples), b—representation of the 106 taxa according to the sample type, c—representation of the 96 samples according to their geomorphological position (Up = upstream riffles, Down = downstream riffles), d—representation of the 106 taxa according to their ecological category, e—representation of the 106 taxa according to their size (meio = meiofauna; macro = macrofauna); Stress = 0.20.

et al., 2003, 2004a; Kibichii et al., 2009]. Other investigations addressed the question of possible contaminations between sampling depths, [Creuzé des Châtelliers and Dole-Olivier, 1990] or the impact of pipe design [Hunt and Stanley, 2000]. Thus, if we currently have a good idea of the limitations of the method, we still do not know exactly its accuracy, especially for the evaluation of invertebrate abundance and density. We found a marked discrepancy between abundance and taxonomic composition obtained from pump-based (BR) versus unbiased samples (BR + BC), with a large range of variation in efficiency. The BR pump underestimates the abundance and diversity of invertebrates but correctly represents the dominant taxa (most frequent and most abundant), indicating an adequate use for rapid assessments of the fauna living in the 0–15 cm zone or where a focus on rare species is not needed.

[31] Finally, even if the pumping techniques are frequently used in the subsurface zone of streambeds [Pennak and Ward, 1986; Bretschko and Christian, 1989; Puig et al., 1990; Boulton et al., 1997, 2004b; Dole-Olivier, 1998; Brunke and Gonser, 1999; Schellenberg et al., 2001; Stubbington et al., 2010a, b, 2011], their efficiency in this particular area of alluvia was still unknown. In this study, the results confirm the underestimation of the abundance and diversity obtained by the BR method, already observed in other investigations via between-techniques comparisons [e.g., Fraser and Williams, 1997; Hahn, 2003]. These biases are not linked to species categories (macro versus meiofauna) or life-style (interstitial versus benthic). In the present protocol, we have neglected the importance of surface fauna, characterized by a particular resistance to tearing, due to its position “above” not “within” the alluvia and the importance of other ecological traits such as morphological resistance to the current force. The next step would be to develop such direct measurement of the accuracy of pumping methods under standard use (i.e., for samples collected deeper into the HZ) with other experimental devices. Undoubtedly, the BR method has many practical advantages and remains sufficiently reliable in most cases. Our results do not question the use of the BR sampling under standard conditions (spatial/temporal comparisons of hyporheic samples), however, being aware of the possible methodological shortcomings and bias is crucial to the interpretation of the data. For example, the fact that BR samples strongly underevaluate density and richness may strengthen the role of the HZ as a refuge, because some studies reported densities of organisms higher in the HZ than in the benthic zone, while BR samples in the HZ are compared to Hess or Surber samples in the benthic zone [e.g., Malard et al., 2001].

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